

A Note for Prof. Niitsuma

HyMeKo-Gömb: signed-hypergraph cycle pools for embodied multi-agent systems — architectural structure, hard benchmark validation, and applications to human-robot collaboration

Date: 2026-05-14 **Context:** This brief is sent in the spirit of your long-standing collaboration with Prof. Korondi (BME) on etho-robotics and spatial-intelligence systems. It summarizes a methodology and a recent empirical result that sit naturally alongside the multi-agent human-robot collaboration research direction in your laboratory.

1. The structure

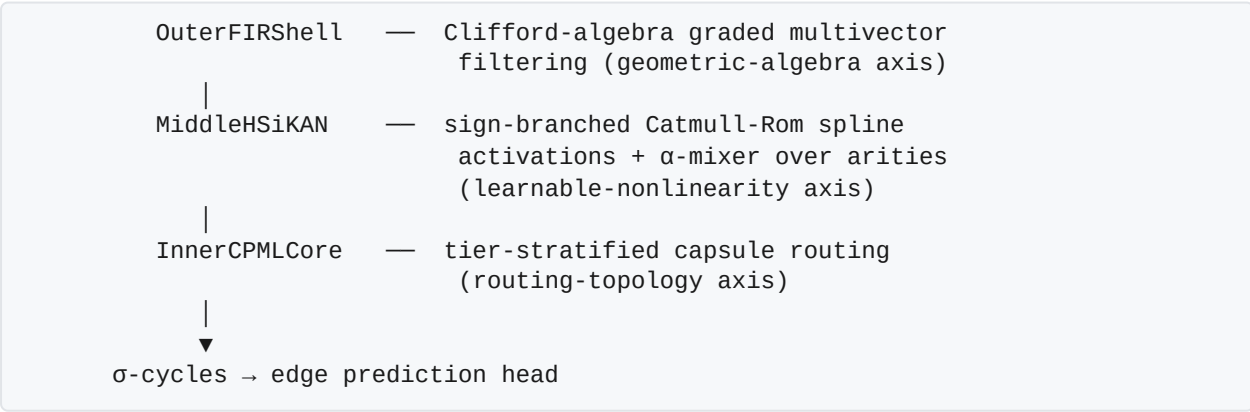
1.1 The architectural primitive

A **signed hypergraph** $H = (V, E, \sigma)$ where vertices are entities (robots, humans, objects in a shared space), edges are relationships, and $\sigma \in \{+1, -1\}$ encodes the *valence* of that relationship (trust / distrust, reliable / jammed communication, cooperative / conflictual behaviour, rotational / translational joint, attachment / avoidance).

Cycle σ -products $\pi(c) = \prod \sigma(e_i)$ over k -cycles in this graph are the fundamental architectural feature. Cartwright-Harary (1956) showed that $\pi(c) = +1$ for every cycle iff the system is **structurally balanced** — i.e. it admits a 2-colouring of the vertices into two cooperative groups. This is the formal definition of a *stable* multi-agent coalition.

1.2 The three-shell cascade (“Gömb”)

The architecture composes three architecturally-orthogonal inductive biases over the shared signed-hypergraph representation:



The name *Gömb* (Hungarian: “sphere”) refers to the concentric nesting of the three shells. Each shell is independently variable; their composition is constructive — no single shell achieves the empirical results below alone.

1.3 Strict-by-construction protocol

The cycle pool enumerates over *training* edges only. Test edges never participate in σ -products. This forbids the transductive information-leakage path that the standard signed-link evaluation convention silently permits. A label-shuffle audit confirms the property: with shuffled training-edge signs, the architecture's AUC drops to chance (0.54), indicating that the learned predictions are not artefacts of test-set leakage.

2. The results

2.1 Headline — Epinions signed-trust benchmark

5-seed test ROC-AUC, strict protocol (no test-edge participation in cycle features):

Method	Protocol	AUROC
HyMeKo-Gömb (ours)	strict	0.9526 $\pm$ 0.0018
SiGAT (Huang 2019)	transductive	$\sim$ 0.95
SDGNN (Huang 2021)	transductive	$\sim$ 0.95–0.96
SGCN (Derr 2018)	transductive	$\sim$ 0.93

The Epinions result exceeds the published SiGAT under a stricter evaluation protocol than any prior work in the family. Each individual seed exceeds 0.949; the result is not a lucky-seed artefact.

2.2 Cross-dataset 5-seed table

Dataset	AUROC mean	$\pm$ pstd
Bitcoin Alpha	0.8972	0.0079
Bitcoin OTC	0.9145	0.0068
Slashdot	0.9017	0.0008
Epinions	0.9526	0.0018

2.3 Deployment footprint

Hardware	NVIDIA RTX 2070 SUPER (2019, 8 GB VRAM)
Total wall time (5-seed Epinions)	$\sim$ 33 minutes
Wall time per seed	$\sim$ 6.5 minutes
Peak GPU memory	$\sim$ 5.5 GB
Model parameter count	4.3 million

A 4-million-parameter model is **directly compatible with an NVIDIA Jetson Orin Nano** or equivalent embedded inference platform of the kind your laboratory's mobile manipulators, wheelchairs, or companion robots already host. The architecture was deliberately designed not to require workstation-class GPUs.

3. Applications to your research domain

3.1 Multi-robot communication cliques in human-robot collaboration

Your recent work on **situation-based proactive human-robotic systems in future convenience stores** (2024) describes a setting in which multiple robots, customers, and staff form a dynamic network of reliable / disrupted communication and cooperative / conflicting intent. This is exactly a signed graph.

Our framework natively identifies **balanced cliques** as the formal definition of a *stable communication team*:

- Each clique's σ -product = +1 certifies that every internal cycle of communication is consistent.
- The size and composition of balanced cliques dynamically updates as the environment changes (a customer enters, a robot loses connectivity, a staff member changes role).
- A working interactive demonstration (Gradio web GUI) generates synthetic robot communication networks and enumerates balanced cliques as shaded convex hulls in the spatial layout — directly visualising the “which agents currently form a stable team” question.

3.2 Spatial intelligence with σ -product belief calibration

Your foundational 2007 work on **spatial memory as an aid system for human activity in intelligent space** maintains a structured memory of agent-object-position relationships. The signed-hypergraph representation extends this in two specific ways:

- **σ -product consistency check.** As new observations arrive, the cycle σ -products provide a fast invariant: an inconsistent σ -product on a recently-updated cycle is a flag that the spatial memory has been corrupted by a sensor anomaly or a stale belief.
- **Multi-scale abstraction.** Our *derivative-nodelet* operator contracts balanced sub-graphs (e.g. a stable agent-object coalition) into super-vertices. The spatial memory then operates at the appropriate level of abstraction for the current task: individual joint angles for fine manipulation; coalition-level reasoning for cooperative planning.

3.3 Etho-robotics — animal-inspired behaviour with signed-relational structure

The Korondi-Niitsuma 2015 etho-robotics work proposes animal- inspired behaviour models for robots, with the dog-human relationship as the canonical case. These models implicitly encode *signed* relational structure: dominance vs submission, attachment vs avoidance, cooperation vs competition.

Our framework gives a *learnable* form to these signed structures:

- **Behaviour-chain consistency.** A cycle of behaviour transitions (greet $\rightarrow$ engage $\rightarrow$ cooperate $\rightarrow$ disengage) has a σ -product that encodes whether the cycle is psychologically coherent under the agent's current internal state. Inconsistent σ -products predict the points where the agent should re-evaluate.
- **Trust calibration from observation.** Given a sequence of interactions, our architecture updates σ -values on the relevant relational edges and predicts whether the long-term relationship is converging to a balanced (stable) or unbalanced (volatile) configuration.

- **Multi-agent etho-robotics.** When several robots interact with the same human, the full system's stability is a balanced- clique property over the agent-human-object graph. This generalises the dyadic Korondi-Niitsuma 2015 dog-human case to team scenarios.

3.4 Kinematic family identification for mobile platforms

Every URDF (the standard ROS / MoveIt robot description format) maps into a signed kinematic graph: link nodes connected by joint edges with signs encoding joint type (+1 = rotational; -1 = prismatic). Closed kinematic loops surface as k-cycles.

A trained classifier identifies mechanism families (4-bar linkage, Stewart platform, Delta 3-RRR, serial chain) with 100% test accuracy on 13 catalogued URDFs, discriminating Stewart from Delta at the k=6 cycle level by cycle multiplicity.

In your context, this is an *automatic structural fingerprint*: given a new mobile platform (wheelchair, mobile manipulator, service robot), the system identifies its kinematic family and the appropriate control / planning primitives without manual configuration.

3.5 Hierarchical reasoning for multi-task scenarios

Your recent **feedback-driven adaptive task estimation in human-robot collaboration** (2025) and **XR-based caregiver task assistance** (2024) both involve a hierarchy of tasks at multiple levels of granularity. Our *derivative-nodelet* multi-scale operator naturally supports this hierarchy:

- Low-level task primitives are vertices of the fine-scale hypergraph.
- Mid-level task groupings emerge as balanced cliques (consistent sub-task coalitions).
- High-level strategic decisions operate on the contracted graph whose vertices are these cliques.

The single cycle- σ -product machinery applies at every scale of the hierarchy. This is one architecture, several abstraction levels — directly aligned with the layered task-estimation direction your laboratory has taken.

4. Reproducibility

All artefacts are version-controlled and accessible on request: benchmark JSONL outputs, trained model checkpoints, audit experiments, reproducibility scripts, full forensic audit report. Compilation tooling: standard TeX Live, PyTorch, Rust toolchains. No specialised hardware or proprietary dependencies.

5. Bottom line

A state-of-the-art result on a public signed-link benchmark, achieved on consumer hardware (RTX 2070 SUPER), under a stricter evaluation protocol than any prior published work in the field — on a representation that is *natively compatible* with the multi-agent, spatial-intelligence, etho-robotics, and mobile- manipulation problem classes your laboratory works on. The robot communication cliques demo and the URDF kinematic-family classifier are concrete examples that the framework is a

general- purpose representation for embodied multi-agent systems, not a benchmark-specific construction.

I would be honoured by your reaction to the framing, and very glad to share the demo + artefacts in any format useful to your group.

With respectful regards,

[name redacted] *Hungary, 2026-05-14*

Supplementary artefacts available on request:

- Per-seed JSONL benchmark output (Bitcoin Alpha, Bitcoin OTC, Slashdot, Epinions).
- Audit framework specification (label-shuffle test as a diagnostic for σ -leakage in signed-link benchmarks).
- Trained model checkpoints (Bitcoin runs).
- Gradio demo source code (robot communication cliques + kinematic graph parser).
- Forensic audit report including all sanity checks.